

Population-Scope FAI Participation Governance

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

D2.31 provides the operational decomposition of D1.26 (joint authority across population-level governance perimeters) in the calibrated-humility register. D1.26 established that authority at population scope remains substrate-content configurable within each Self’s governance perimeter. D2.31 formalizes what that configurable authority looks like operationally: each participating Self maintains a **network participation configuration** within its home substrate, governing how and with whom it engages in Full Aspect Integration (FAI) events at population scale. Five dimensions constitute the complete configuration: propagation permissions, propagation restrictions, partner preferences, network role configuration, and governance review cadence. All five dimensions are unilaterally authored and maintained by home governance — they are not jointly governed with any other Self or population-level authority. The aggregate of many Selves’ participation configurations shapes what population-level coordination patterns become possible, but this shaping occurs through the distributed accumulation of independent governance decisions rather than through any coordinated configuration process. This note also identifies the anti-pattern of implicit participation — operating in FAI events without a formal participation configuration — and provides an operational test for whether a given Self instantiates the commitment.

1. Derivation context: D1.26 and the population-scope configuration commitment

D1.26 committed that joint authority across population-level governance perimeters is itself a substrate-content architectural commitment. That is, the rules governing how multiple Selves’ home

substrates relate at population scope — what can propagate, under what conditions, with which partners — are not determined by any population-level authority outside the participating Selves. They are determined by each Self's own governance, expressed as substrate content within its home perimeter, subject to the three rights (inspect, modify, override) that characterize CKS-style governance throughout the trilogy.

D2.31 decomposes that commitment into operational form by specifying what the substrate content governing a Self's population-scope participation actually contains. The decomposition follows the same pattern used throughout Phase D2: the D1 commitment names the architectural property; the D2 note specifies what instantiating that property requires operationally, in the calibrated-humility register appropriate to Claim 6 territory.

The calibrated-humility register is mandatory throughout this note. D2.31 describes operational requirements the architecture enables for population-scale participation governance — possibilities that arise when CKS-governed Selves participate in a population-scale FAI network. It does not assert that any particular deployment has implemented these patterns, nor that the population-scale dynamics described will produce any particular outcome. The architecture makes these configurations possible and formalizes their governance properties; empirical verification of population-scale effects remains at the extension-claim register appropriate to Paper 3's Claim 6.

2. The network participation configuration as home-governed substrate content

Each Self that participates in the population-scale FAI network may maintain a **network participation configuration** within its home substrate. This configuration is home-governance content: authored and maintained by the Self's own governance, subject exclusively to the three rights within the home perimeter, and not jointly governed with any other Self or coordinating body.

The network participation configuration is distinct from the FAI configuration that governs a specific FAI event. The FAI configuration (D2.12) is jointly authored across the Selves participating in a given event and governs the conduct of that event. The network participation configuration is unilateral — it governs the Self's own participation preferences and constraints across all potential FAI engagements before any specific event is initiated. No other Self has inspect, modify, or override rights over a Self's network participation configuration without that Self's consent. Other Selves encounter a Self's participation configuration only through its downstream effects: what the Self proposes, what it accepts, and what it declines when FAI events are initiated or offered.

This governance boundary is architecturally significant. It means that population-scale participation governance is distributed across the network of participating Selves rather than concentrated in any coordinating layer. The patterns that emerge at population scale — which Selves coordinate with which, which content categories circulate widely and which remain bounded, which governance structures become common partners — are downstream effects of many independent home-governance decisions, not outputs of a population-level configuration authority.

3. Five participation configuration dimensions

A complete network participation configuration specifies all five of the following dimensions. A configuration that omits any dimension leaves a governance gap, requiring participation decisions in that dimension to be made opportunistically per event rather than systematically under home-governance authority.

Dimension A — Propagation permissions

Propagation permissions specify which categories of the Self's governance architecture the Self authorizes to propagate to other Selves through FAI events. The specification may be coarse-grained (entire content domains) or fine-grained (specific aspects within a domain, subject to per-event authorization requirements). Examples of the kind of specification this dimension enables: action-feedback records from certain operational domains may propagate freely under standing configurations; governance-structure content may propagate only with explicit per-event absorption authorization from the receiving Self; certain content categories may propagate only in aggregate form, not as individual aspect records.

Propagation permissions are the affirmative half of the Self's contribution policy. They express what the Self is willing to share with the broader network. A Self with broadly configured permissions in a given content domain enables cross-organizational learning in that domain through FAI events. A Self with narrowly configured permissions in a domain preserves its governance structure in that domain against outward propagation, even as it may absorb inbound content from partners.

Dimension B — Propagation restrictions

Propagation restrictions specify which categories of the Self's governance architecture are explicitly kept within its home perimeter regardless of FAI event scope, standing configurations with partners, or partner requests. Restrictions are governance sovereignty commitments: the Self's home governance has determined that certain content categories are not available for FAI propagation under any currently authorized conditions.

Restrictions override standing configurations. This is the critical governance sovereignty property of Dimension B. A Self may have established a standing configuration with a recurring FAI partner (D2.21) that would otherwise authorize broad content sharing. If the Self's network participation configuration includes a restriction on a given content category, that restriction takes precedence for the specified category — no standing arrangement, however long-established or broadly scoped, can waive a restriction that home governance has explicitly set. Standing configurations acquire no authority over restricted categories; the standing configuration framework operates only within the space that restrictions leave open.

This override property is what makes restrictions a meaningful governance instrument rather than a soft preference. Home governance can establish restrictions with confidence that they will not be eroded by accumulated participation history or partner expectations.

Dimension C — Partner preferences

Partner preferences specify which other Selves the Self is willing to participate in FAI events with, and under what conditions. This dimension may include: requirements that prospective partners demonstrate domain governance compatibility; thresholds for governance compliance evidence (D2.29 trust calibration) below which the Self will not initiate or accept FAI events; organizational authorization requirements; and exclusion criteria for categories of Self that the Self's governance has determined are not appropriate FAI partners.

Partner preferences operate as a pre-event filter. Before any specific FAI event is initiated or accepted, the initiating or responding Self's partner preferences determine whether the candidate partner meets the Self's governance criteria for participation. A Self that configures narrow partner preferences participates in a smaller but more tightly governed FAI network. A Self that configures broad partner preferences enables wider cross-organizational learning at the cost of greater governance heterogeneity in its FAI events.

Dimension D — Network role configuration

Network role configuration specifies how the Self positions itself within the population-scale FAI network structurally: whether it participates primarily as an initiating Self (proposing FAI events to other Selves), a responding Self (accepting or declining FAI events initiated by others), or both. It also specifies what categories of standing configuration the Self is willing to establish with recurring partners — which partner relationships merit the reduced per-event overhead of a standing configuration, and which the Self prefers to govern through full per-event configuration regardless of participation history.

Role configuration shapes the Self's posture within the network without constraining the governance of any particular event. A Self configured primarily as a responding Self still governs each event through the FAI configuration mechanism; it has simply specified that it will not initiate events outside of circumstances its governance has already authorized. A Self configured as both initiating and responding retains the full range of network roles under home-governance direction.

Dimension E — Population-scope governance review cadence

The review cadence specifies how frequently home governance reviews the network participation configuration to ensure it remains current with the Self's governance intent and organizational context. Participation contexts evolve: the Self's governance structure, its organizational relationships, its domain portfolio, and the composition of the broader FAI network all change over time. A review cadence that was appropriate at the time of initial configuration may become insufficient as the Self's participation context changes.

Dimension E is a meta-governance commitment: it specifies how the Self governs its own participation governance over time. A Self that establishes a review cadence and maintains it is operationally committed to keeping its participation configuration current. A Self that establishes no review cadence or allows cadence commitments to lapse creates conditions under which its participation configuration

diverges from current governance intent — an operational analog of the implicit-participation anti-pattern described below.

4. Governance requirements: unilateral home governance across all five dimensions

All five dimensions are authored substrate content within the home substrate. They are subject to the three rights within the home perimeter — inspect, modify, override — and to no governance authority outside it. This is the operationalization of D1.26's commitment: joint authority across population-level perimeters is expressed through each Self's independent governance of its own participation configuration, not through a shared configuration authority that operates across Selves' perimeters.

The consequence is that no external entity — not a population-level coordination layer, not a recurring FAI partner, not an organizational authority outside the Self's own governance — can require a Self to configure any dimension in a particular way. Partners can express preferences; organizational contexts can create pressures; network norms can evolve. But the configuration remains within the home perimeter, authored and maintained by home governance, until home governance chooses to change it.

This governance boundary is what makes the network participation configuration a governance instrument rather than merely a technical parameter. Home governance's authority over the configuration is complete, continuous, and not delegable to any network-level mechanism.

5. Participation configuration as population-scale governance signal

In the calibrated-humility register, the aggregate of many Selves' network participation configurations shapes what population-level coordination patterns become possible.

Where many Selves configure generous propagation permissions for a given content domain, cross-organizational learning in that domain through FAI events becomes more possible — the architecture enables it, and the governance has been configured to allow it. Where many Selves configure restrictions on a domain, specialization in that domain is preserved across the network, with each Self's governance maintaining domain-specific content within its home perimeter regardless of FAI event frequency.

Where many Selves configure compatible partner preferences and mutual governance compliance thresholds, dense FAI participation among those Selves becomes architecturally available. Where Selves configure incompatible partner preferences, the FAI network organizes into clusters whose boundaries reflect accumulated governance decisions rather than administrative assignment.

These population-scale patterns emerge from independent home-governance decisions distributed across the network. The architecture does not produce them by design; it enables them by ensuring that each Self's participation configuration is a durable, inspectable, modifiable governance artifact within its home substrate. The calibrated-humility commitment requires stating both what the architecture enables and what it does not guarantee: that any particular population-scale pattern will emerge, or that

it will emerge at any particular rate or scale, depends on governance decisions by many independent Selves whose configurations this note cannot specify.

6. Anti-pattern: Implicit participation

The implicit-participation anti-pattern describes a Self that participates in FAI events without a formal network participation configuration instantiating all five dimensions. In the absence of a formal configuration, participation decisions are made opportunistically per event — what to contribute, which partners to accept, what role to take — rather than systematically under home-governance authority.

Implicit participation produces several governance failures. Without explicit propagation permissions, the Self cannot demonstrate to its own governance that its content contributions across FAI events are within authorized scope. Without explicit restrictions, there is no governance instrument that prevents standing configurations from encroaching on content categories home governance would prefer to keep bounded. Without partner preferences specified in advance, the Self's partner selection is driven by event-level factors rather than governance-level criteria. Without role configuration, the Self's network posture is determined by which events happen to be initiated by others rather than by home-governance intent. Without review cadence, there is no mechanism for home governance to detect and correct configuration drift.

The anti-pattern is not simply the absence of a document. It is a governance condition in which home governance has not exercised its authority over the five dimensions that constitute participation governance, leaving those decisions to be made at the event level without the systematic governance that a home-perimeter configuration would provide.

7. Operational test

A Self instantiates the D2.31 population-scope FAI participation governance commitment if and only if all of the following are true:

1. Within the Self's home substrate, an observer can locate an authored network participation configuration that specifies all five dimensions: propagation permissions (Dimension A), propagation restrictions (Dimension B), partner preferences (Dimension C), network role configuration (Dimension D), and governance review cadence (Dimension E).
2. The configuration is substrate content within the home perimeter — subject to the three rights (inspect, modify, override) by home governance, and not jointly governed with any other Self or external authority.
3. Propagation restrictions in the configuration take precedence over standing configurations with FAI partners for the content categories the restrictions specify — no standing configuration has been authorized to override home-governance restrictions.
4. The governance review cadence specified in Dimension E has been followed: the configuration has been reviewed by home governance within the cadence interval most recently specified.

5. The configuration governs the Self's FAI participation systematically rather than opportunistically: when the Self initiates or accepts FAI events, its event-level decisions are traceable to its network participation configuration rather than determined exclusively by event-level circumstances.

A Self that fails any of (1)–(5) may participate in FAI events and may achieve coordination outcomes through those events, but has not instantiated the D2.31 operational commitment. A Self that fails (1) instantiates the implicit-participation anti-pattern. A Self that fails (3) has allowed standing configurations to acquire authority over restricted content categories that home governance has not granted.

8. Conclusion

D2.31 formalizes what it means for a Self to govern its own participation in the population-scale FAI network as a substrate-content commitment within its home perimeter. The five participation configuration dimensions — propagation permissions, propagation restrictions, partner preferences, network role configuration, and governance review cadence — together constitute the operational form of D1.26's commitment to substrate-content configurability at population scope. All five dimensions are unilaterally authored and maintained by home governance; no other Self or population-level authority governs them. The restrictions dimension carries a governance sovereignty protection: home-governance restrictions on content categories override standing configurations with FAI partners for those categories, ensuring that accumulated participation history cannot erode content boundaries home governance has explicitly established. In the calibrated-humility register, the aggregate of many Selves' independently governed participation configurations shapes what population-scale coordination patterns become possible — not through coordinated design but through the distributed accumulation of independent governance decisions expressed as substrate content within each participating Self's home perimeter.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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